

Assignment 7: The Great Firewall of Santa Cruz - Bloom Filters, Linked Lists, Binary Trees and Hash Tables DESIGN.pdf

Files to be included in directory “asgn7” + pseudocode/structure:

- banhammer.c: This contains main() and may contain the other functions necessary to complete the assignment.
- messages.h: Defines the mixspeak, badspeak, and goodspeak messages that are used in banhammer.c. Do not modify this.
- salts.h: Defines the primary, secondary, and tertiary salts to be used in your Bloom filter implementation. Also defines the salt used by the hash table in your hash table implementation.
- speck.h: Defines the interface for the hash function using the SPECK cipher. Do not modify this.
- speck.c: Contains the implementation of the hash function using the SPECK cipher. Do not modify this.
- ht.h: Defines the interface for the hash table ADT. Do not modify this.
- ht.c: Contains the implementation of the hash table ADT.
- bst.h: Defines the interface for the binary search tree ADT. Do not modify this.
- bst.c: Contains the implementation of the binary search tree ADT.
- node.h: Defines the interface for the node ADT. Do not modify this.
- node.c: Contains the implementation of the node ADT.
- bf.h: Defines the interface for the Bloom filter ADT. Do not modify this.
- bf.c: Contains the implementation of the Bloom filter ADT.

- `bv.h`: Defines the interface for the bit vector ADT. Do not modify this.
- `bv.c`: Contains the implementation of the bit vector ADT.
- `parser.h`: Defines the interface for the regex parsing module. Do not modify this.
- `parser.c`: Contains the implementation of the regex parsing module.
- `Makefile`
 - `CC = clang` must be specified.
 - `CFLAGS = -Wall -Wextra -Werror -Wpedantic` must be specified.
 - `pkg-config` to locate compilation and include flags for the GMP library must be used.
 - `make` must build the `encrypt`, `decrypt`, and `keygen` executables, as should `make all`.
 - `make decrypt` should build only the `decrypt` program.
 - `make encrypt` should build only the `encrypt` program.
 - `make keygen` should build only the `keygen` program.
 - `make clean` must remove all files that are compiler generated.
 - `make format` should format all your source code, including the header files.
- `README.md`
 - In proper Markdown syntax.
 - Describe how to use your program and `Makefile`
 - Lists and explains any command-line options that program accepts.
 - Reports any false positives by `scan-build`

- Notes down any known bugs or errors in this file as well for the graders.
- WRITEUP.pdf: This document must be a PDF. The writeup must include at least the following:
 - Graphs comparing the total number of lookups and average binary search tree branches traversed as you vary the hash table and Bloom filter size.
 - How do the heights of the binary search trees change?
 - What are some factors that can change the height of a binary search tree?
 - How does changing the Bloom filter size affect the number of lookups performed in the hash table?
 - Analysis of the graphs you produce.
- DESIGN.pdf
 - This document must be a proper PDF.
 - Describes design and design process for program with enough detail such that a sufficiently knowledgeable programmer would be able to replicate implementation.